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Role of Human-Computer Interaction Factors as Moderators of Occupational Stress and Work Exhaustion

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Software professionals perform boundary-spanning activities, and thus need strong interpersonal, technical, and organizational knowledge to be professionally competent. They have to perform in a demanding work environment characterized by strict deadlines, differing time zones, interdependency in teams, increased interaction with clients, and extended work hours. These characteristics lead to occupational stress and work exhaustion. Yet, the impact of stress is felt in different ways by different people, even if they perform the same functions. These differences in the perception of stress can be caused by varying confidence in their technical capabilities. People possess varying technical capabilities, based on their acquisition of technical skills, comfort level in using the technology, and intrinsic motivation. These attributes represent the human-computer interaction (HCI) personality of software professionals. This article examines whether these HCI factors moderate the relationship between occupational stress and work exhaustion. Data were collected from software professionals located in Chennai and Bangalore in India. The data revealed that HCI factors had a main effect but no significant moderating effects on work exhaustion. The control over the technology variable emerged as the key variable among the HCI factors that affected software professionals' ability to cope with stress and work exhaustion.

1. INTRODUCTION

People need to possess competencies that enable organizations to compete at the forefront of the technology curve (ITAA, 1999). Software professionals perform

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various roles in the process of software development (Rajeswari & Anantharaman, 2003). Hence, they need strong interpersonal, technical, and organizational knowledge to be professionally competent. The technical functions of software professionals include requirements analysis, design, development, testing, and delivery. In addition, software professionals are involved in activities such as technology appraisal; design; development; and delivery of software, people management, and resource management functions. This pattern of functions is similar across the country and across organizations.

Despite performing similar functions, some professionals are able to cope with the pressures of being a software professional and continue to contribute effectively to the organization as well as the profession, whereas many others are not. These differences also are reflected in the number of hours they spend on the actual job, their depth and variety of knowledge, their confidence in use of software technologies, and their innovative use of software. These activities involve human-computer interaction (HCI). People vary in their HCI behavior as much as they vary in their personality. A number of studies (Jex & Bliese, 1999; Jex & Gudanowski, 1992; Schaubroeck, Lam, & Xie, 2000) have explored the role of individual differences in personality (e.g., self-efficacy beliefs, collective efficacy beliefs) as moderators of occupational stress. Similarly, some recent studies have explored the role of computer self-efficacy (CSE) as a moderator in the context of burnout (Salanova, Grau, Cifre, & Llorens, 2000; Salanova, Peiro, & Schaufeli 2002) among information technology users. In a similar vein, it is pertinent to explore whether HCI factors such as CSE moderate the relationship between stress and work exhaustion, particularly among software professionals, because it is a technology-oriented and knowledge-intensive profession. Moreover, occupational stress is an area of serious concern among software professionals, because 95% of senior managers, 90% of middle managers, and 80% of the rank and file in the information systems environment describe their day as stressful (Garner, 1997). The situation in India is not different. A chronic shortage of skilled professionals, increased workload, difficulties in planning for projects, tighter deadlines because of competition, and juggling of tasks contribute to the stress of software professionals in India (Natarajan, 2000). Moreover, stress is identified as a precursor to turnover (Moore, 2000). Therefore, the need to address these issues and look for solutions from yet another perspective, HCI, assumes greater significance.

The focus of this study, therefore, is to inquire whether HCI factors play a moderating role in the relationship between occupational stress and work exhaustion. This article (a) deals with the theoretical background on issues relating to HCI, the existing measures of HCI, the development of scales to measure various dimensions of HCI relevant to the context of the study, and the data analysis pertaining to scale development and (b) investigates the moderating effects of HCI variables in the relationship between occupational stress and work exhaustion.

2. THEORETICAL BACKGROUND

Several studies have pointed out that individual differences, particularly individual talent and skill, predict project performance (Curtis, Krasner, & Iscoe, 1988;

Rasch, 1989). The key to organizational success lies in identifying such talent and ability among software professionals, enabling professionals to perform well, and working toward the retention of such professionals. Retention of software professionals is dependent on creating an environment that increases favorable factors at work and decreases unfavorable conditions. Occupational stress is identified as one of the unfavorable factors that influence intent to turnover (Moore, 2000) and, therefore, that needs to be addressed in the context of retention of key resources.

The major stressors for software professionals are fear of obsolescence, individual team interaction, client interaction, work–family interface, role overload, work culture, technical constraints, lack of family support, workload, and technical risk propensity (Rajeswari & Anantharaman, 2003). A careful analysis reveals that sources of stress exist across two basic dimensions—one dimension deals with soft skills and people competencies and the other dimension deals with software skills and technical competencies. But they are interlinked with each other as stress permeates from one dimension to another and vice-versa. Repercussions of stress felt by people are manifold and have spillover effects.

Further analysis of stress related to technology usage reveals that fear of obsolescence is a key source of stress. Hence "keeping up" with emerging IT has been identified as a critical issue in Canada (Carey, 1992) and the United States (Schambach & Blanton, 2002). Technical constraints on the job and technical risk propensity are also stressors related to the use of technology.

The perception of a situation as a stimulus or a stressor is based on the appraisal of the situation as potentially threatening. Lazarus (1966) observed that, "The appraisal of threat is not a simple perception of the elements of the situation, but a judgment, an inference in which the data are assembled to a constellation of ideas and expectations." The perception of stress by an individual varies according to the way the individual appraises the situation that triggers stress. The appraisal varies according to the "vulnerability profile"—personality, demographic factors, physical make-up, past experience, and motivation (Appley & Trumbull, 1967)—of the individual. In the context of software professionals, vulnerability profile may include confidence in using a particular technology, platform, or operating system; control over technology; and such other factors that constitute human interaction factors. In the context of software development professionals, who constantly use technology, particularly computers, factors pertaining to HCI play a key role in the appraisal of work situations as stressful and nonstressful.

Therefore, it is possible that individual variations in HCI factors such as beliefs in computer skills (computer self-efficacy), intrinsic motivation, and training efficacy play a key role in enabling software professionals to handle stress at work in general and stress particularly from fear of obsolescence, technical constraints, and technical risk propensity.

The literature on information systems professionals points out that consequences of stress among information systems professionals include work exhaustion, intent to turnover (Moore, 2000), lack of organizational commitment, and poor job satisfaction (Igbaria & Greenhaus, 1992). Moreover, the literature indicates that stressors such as role ambiguity and workload lead to work exhaustion and intent to turnover (Moore, 1998). In addition, turnover is related to burnout (Weisburg, 1994), of which emotional work exhaustion is a component. Work exhaustion has been found to be prevalent among software professionals (Moore, 1998, 2000; Rubin & Hernandez, 1988) and in particular among software developers in the software industry (Sonnentag, Brodbeck, Heinbokel, & Stolte, 1994). Symptoms related to work exhaustion—such as low energy, feeling restless, and inability to concentrate—that are associated with the presence of work exhaustion have been found to be present among software professionals (Weiss, 1983). Hence, work exhaustion was chosen as an outcome variable in this study. Work exhaustion construct refers to the depletion of emotional and mental energy needed to meet job demands (Moore, 2000). It was assessed using the work exhaustion subscale of MBI-GS (Schaufeli et al., 1996), which has five items.

2.1. CSE

CSE reflects an individual's perception of his or her ability to use computers in the accomplishment of a task (Compeau & Higgins, 1995b). Increased usage of computers and software enhances an individual's familiarity with the system and the confidence of the individual in using the computer or software. This concept is brought out in the construct of CSE. In the context of this study, the task refers to software development activity. Researchers have explored the nature of CSE to understand its antecedents (encouragement, experience, and management support) and consequences (outcome expectations, anxiety, affect, and actual performance). These studies are mentioned briefly in Marakas, Yi, and Johnson (1998). The common thread among these studies is that attitude determines behavior. In other words, attitude toward the language, platform, and operating systems determines the performance using the language, platform, and operating systems. Extracting further from this concept in the context of software professionals, one can say that CSE is a trait determined by the combination of the ability to use technology to adapt to changes in technology and the competency to deal with ambiguous situations involving the use of technology.

Perception of ability to perform specific computer-related tasks in the domain of general computing is known as task-specific CSE. An individual's judgment of efficacy across multiple computer application domains is known as general CSE. For example, the end users who use a personal computer for individual usage can interpret the general CSE as applicable to all tasks that he or she does on a daily basis using a computer. However, for software professionals, with an appropriate background, general CSE may mean the efficacy of using computers for the purpose of software development across multiple domains, platforms, technologies, and application environments. General CSE is more a product of a lifetime of related experiences and tends to more closely confirm the definition of CSE that is often offered and tested in the information systems literature (Martocchio, 1994).

Two recent studies used the construct of CSE as a moderator in the context of burnout among information technology users (Salanova et al., 2000, 2002). The first study examined the role of CSE in the relationship between computer training, frequency of usage, and burnout. The second study examined the role of self-efficacy and CSE as moderators in the demand control model of Karasek (1979), with burnout as the dependent variable, where demands were measured by quantitative workload, and control was measured using time and method control. There have been no such studies of this nature in the context of software development professionals.

A review of the studies on CSE literature and the meta-analysis of Marakas et al. (1998) also revealed that CSE has not been examined among the group of software professionals. This is probably because software development professionals are assumed to have more CSE than the normal computer user. Their choice of the software development profession also reveals that their CSE is high. Because the context of CSE is a dynamic trait and is context and situation specific, this group of professionals need to be examined as they face the greater threat of obsolescence more than any other information technology user because of continuous changes in technology. In addition, they are most likely to be the front runners in the use of latest techniques, languages, platforms, and operating systems because of the demands of the clients or end users. This creates ambiguity and complexity in the task profile of software professionals. Because task characteristics such as ambiguity and complexity affect perceptions of efficacy, software development professionals are more susceptible to change in CSE. This may in turn determine the way they appraise a computer- or software-related problems at work as stressful or otherwise. If their lack of CSE leads them to appraise a situation as stressful, then such software development professionals would be more prone to occupational stress, which would affect performance. Hence, there is a need to explore the role of CSE among software development professionals in the context of occupational stress.

CSE has three interrelated but distinct dimensions: strength, magnitude, and generalizability. There are several scales to measure CSE (Marakas et al., 1998). Of these scales, the one developed by Compeau and Higgins (1995b) was found to be most suitable. Because that scale is directed toward novice users, the preface to the instrument was modified to address the context of experienced software professionals. In addition, two more items that depict various situations (typically expected to be present in software companies) were added to the existing 10 items. They depict confidence in completing a program or design using a new software technology, package, or language: Someone giving step-by-step instructions and excelling at the use of the new technology. The resultant scale with 12 items was used in this study.

2.2. Intrinsic Motivation

Intrinsic motivation refers to the pleasure and inherent satisfaction derived from a specific activity (Vallerand, 2000). Intrinsic motivation is defined as the desire to work on a task for its own sake, because the work itself is enjoyable, satisfying, and or challenging. In the context of computer usage, intrinsic motivation has been measured using the construct of computer playfulness (Venkatesh, 2000). The characteristics of an individual with playful dispositions are described as follows (Webster & Martocchio, 1992):

> Individuals with playful dispositions are said to be guided by internal motivation, an orientation toward process with self-imposed goals, a tendency to attribute their own meanings to objects or behaviors (that is, to not be dominated by a stimulus), a focus on

> pretense and nonliterality, a freedom from externally imposed rules, and active involvement. (p. 202)

Playfulness is a manifestation of cognitive spontaneity, which reveals itself as curiosity and inventiveness. In the context of HCI, this trait enables an individual to try out different patterns of behavior of the software technology or package under various conditions. This familiarity gets transformed into mastery with great ease (in course of time and with practice). To denote this process, Webster and Martocchio (1992) developed the construct called "micro computer playfulness." It is defined as "the degree of cognitive spontaneity in microcomputer (in the context of this study computer) interactions" (p. 202).

Computer playfulness has been studied typically in the context of Theory of Planned Behavior (TPB) and Technology Acceptance Model (TAM) as a predictor for ease of use and intention to use in the case of both management information systems professionals and end users. It is found to be positively related to computer efficacy beliefs and training outcomes (Webster & Martocchio, 1992). Hence it is expected that intrinsic motivation also will be a concomitant moderator of occupational stress in this study. The 7-item computer playfulness scale of Webster and Martocchio (1992) has been used to measure intrinsic motivation.

2.3. Technology-Training Efficacy

Efficacy beliefs with respect to training have been identified as training efficacy beliefs and posttraining efficacy beliefs. Research has demonstrated that people with high CSE beliefs perform better on declarative knowledge tests after training (Martocchio & Judge, 1997; Webster & Martocchio, 1992). Relationships between training experience and posttraining efficacy have been explored during flight training (Sherman, 1997). Perceived knowledge about a specific task or about an environment is expected to influence one's self-efficacy. The results of the study by Torkzadeh et al. (1999) indicated a positive relationship between training and CSE. A linear relationship was found between training and ability (Nelson, 1990) because success of the training depends on a person's ability to benefit from training, which includes acquisition of knowledge and skills as well as utilization of previous experiences (Cronbach & Snow, 1977). Because training is positively related to CSE (Torkzadeh et al., 1999), it may act as a moderator in the relationship between stress and its consequences in the case of software development professionals. However, there has not been an attempt to capture the aggregate of a person's efficacy that has been accumulated through training (particularly with respect to computing technology) over a period. The efficacy that people have in the technology training that they have accumulated can be defined as technology-training efficacy. To measure the technology-training efficacy, an instrument needs to be developed and used in this study.

In the context of software professionals, technology training that is continuous in nature equips them with the latest skills in the field of computing. Training includes formal classroom training as well informal training and self-learning. Existing scales measure the training efficacy of software professionals on a particular technology, operating system, or language at the end of a training program. These are not adequate to measure the confidence in the training that software professionals have accumulated over the years. The level of confidence in the training of software professionals is reflected in the belief that people have in their ability to transfer their knowledge to others through training. Research regarding self-efficacy judgments about computers has focused on novice computer users and introductory applications (Potosky, 2002). Few studies involve training and efficacy judgments over complex computer-related applications and technology usage, as in the process of software development. Hence, a 3-item scale was developed in the study to measure technology-training efficacy.

Several studies have shown that training increases self-efficacy and that self-efficacy is related to performance (Gist, 1989; Gist, Schwoerer, & Rosen, 1989; Martocchio & Webster, 1992; Saks, 1994). In addition, self-efficacy is related to technology acceptance (Gist, et al., 1989; Hill, Smith, & Mann, 1987; Martocchio & Webster, 1992) and performance in software training sessions (Compeau & Higgins, 1995a; Gist et al., 1989; Webster & Martocchio, 1992;). Because self-efficacy moderates the impact of occupational stress, it is expected that technology-training efficacy can act as moderator of occupational stress, especially among software professionals.

2.3. Perceived Control Over Technology

Perceived behavioral control is defined as " … the individual's perception of the ease or difficulty of performing the behavior of interest" (Ajzen, 1991, p. 183). Taylor and Todd (1995) related this construct to the perceived difficulty in technology usage. The ease or difficulty of performing a behavior is a function of the knowledge or skills possessed by a person. It is also dependent on the availability of resources at the disposal of the individual, in addition to the autonomy of choice, regarding software technology that can be used most successfully to achieve a goal. It is often used as an anchor in the context of Theory of Planned Behavior (Ajzen, 1991).

Perceived behavior control refers to individual's perceptions regarding their capability to perform a given behavior based on the extent to which they have requisite knowledge, resources, and control at work. Perception of control over technology of software professionals can be measured in terms of the choice of technology usage they possess and access to technical resources (in terms of infrastructure for learning). It is defined as a person's perception of the extent to which resources and skills required (to make use of technology) are actually available (Yi & Venkatesh, 1996).

In the context of this study, perceived behavior control is used in the specific context of software professionals to determine control over technology. The measure of control over technology has been used to measure the autonomy related to technology usage available to the individual in the work environment. In general, perception of control also reflects the autonomy in the organizational environment in relation to the use of a specific system (Venkatesh, 2000). This measure has been enhanced to encompass control over technical resource availability and usage.

A person's perceived lack of control over their environment (work situation in general and work using computers, in particular) contributes to the perception of stress (Wingreen & Blanton, 2001). Therefore, it becomes important to study the role of these variables, namely, perceived control on the job and perceived control over technology as moderators in the stress–strain process in the context of software development professionals.

3. MOTIVATION FOR THIS WORK

- Perception of stress is based on a person's appraisal of a situation as stressful or not. Therefore, someone who is good in computing in "C" language is likely to consider working on "C" as less stressful as compared with someone who feels he or she is not so good in computing in "C" language. In this context, CSE may play a significant role in moderating the stress of software professionals.
- Confidence in the training that software professionals have accumulated in various technologies enables them to work in various platforms or technologies. However, this measure has not received attention as much as posttraining efficacy tests, although in the context of software professionals, many times training may be only in a given area, and software professionals may be asked to work in other similar technologies without further training. It is this technology-training efficacy that enables them to function well. Therefore, there is a need to develop a scale for technology-training efficacy and to explore and understand its moderating role in the context of occupational stress and work exhaustion.
- Technology acceptance influences decisions to learn a particular technology, to use it, and to master it. Intrinsic motivation using computer playfulness is a key determinant of technology acceptance (Venkatesh, 2000). Therefore, if software professionals are intrinsically motivated to use a technology for computing, it is likely that their appraisal of the situation as stressful is less likely to occur, and, hence, intrinsic motivation can be a moderator of the context of occupational stress.
- Moreover, these variables have not been concomitantly examined as moderators of occupational stress.

4. HYPOTHESES

The main objective of this article is to explore the concomitant role of the HCI variables such as CSE, technology-training efficacy, perceived control over technology, and intrinsic motivation as moderators in the relationship between occupational stress and work exhaustion among software professionals.

The other objectives are as follows:

1. to modify the scale of CSE developed by Compeau and Higgins (1995b) to suit the current context of the study and to determine the reliability and validity of the CSE scale
2. to develop a scale for technology-training efficacy and determine its reliability and validity

3. to develop a scale for perceived control over technology
4. to check the reliability and unidimensionality of the scale for intrinsic motivation (computer playfulness) in the context of this study
5. to examine the relationship between stress, HCI variables, and outcome variables

The following hypotheses were framed to examine the moderating role of HCI variables in the relationship between occupational stress and work exhaustion:

H1a. Stress is negatively associated with all HCI variables.
H2a. Stress and HCI variables have main (additive) effects on work exhaustion.
H2b. Stress and HCI variables have an interaction effect on work exhaustion.

5. INDIAN SOFTWARE INDUSTRY—AN OVERVIEW

The Indian software industry is a small but growing part of the global software industry. India currently exports to 95 countries around the globe, and 185 of the fortune 500 companies have outsourced some of their software requirements to India. The world has recognized India's competitive advantage in software services, and today India is a magnet for software clients owing to the quality of its skilled workforce (NASSCOM, 2001). Several multinational companies are attracted to the Indian software market because of its enormous pool of skilled professionals who are proficient in the English language. India has been identified as a top destination for software outsourcing by 82% of the American companies because of the technical excellence of Indian software professionals according to a discussion article from Harvard University (Bajpai & Rajdou, 1999).

The software industry consists of organizations that have a flat hierarchy, young management teams, and informal but professional management styles with an emphasis on punctuality and efficiency (Arora, Arunachalam, Asundi, & Fernandes, 1999). Human resources are not only the value creators and drivers in the Indian software industry but also the intellectual infrastructure of this knowledge intensive industry. Therefore, attracting, training, retaining, and motivating existing employees are key determinants of success in the Indian software industry. Empowering, engaging, and energizing employees are established ways of creating effectiveness and efficiency through motivation in the Indian software industry (Karnik, 2003). A significant challenge the industry faces is in the creation of great leaders who are capable of steering the organizations and their employees toward greater success.

Retention of employees is another significant challenge in the Indian software industry. Stressors in the workplace affect the well-being of people by causing negative moods (Weiss, 1983) and burnout (Moore, 2000). In this process, the well-being of the organization is affected because of low productivity, increased costs in recruitment and retraining of professionals, adjustment problems at work, and lack of availability of skilled resource persons. Therefore, addressing the issue of stress at work is vital to the sustained growth and development of Indian software industry.

6. SCALE DEVELOPMENT

6.1. Modification of the CSE Scale

The CSE scale developed by Compeau and Higgins (1995b) is task focused. It incorporates elements of task difficulty and captured differences in self-efficacy magnitude as well (Compeau & Higgins, 1995b). Even though it was found to capture all the aspects concerning use of computing technologies, it has failed to cover multiple software applications used in the work environment of software professionals. Hence, the preface of the scale was modified to include various types of software technologies. This scale was modified to include items that measure the confidence to perform when step-by-step assistance from someone is available and to measure the confidence in achieving excellence in the use of a new software package for software development. The respondents have been asked to respond to their level of confidence in the use of a new software, technology, or package during the course of software project development and indicate the same on a 7-point scale, where 1 = *not at all confident* and 7 = *absolutely confident*.

6.2. Development of the Technology-Training Efficacy Scale

Three items have been framed to measure the technology-training efficacy scale. They include the ability to train oneself, which reflects the confidence that one has already accumulated a good training. The second item reflects the readiness and interest to receive further training. The third item reflects the confidence of software professionals to impart training to others. It is difficult to train others when one is not sure of oneself. This item, therefore, measures the confidence that one has acquired and conveys the readiness to impart training to others. Face validity and content validity of the instrument has been established using the opinions of experts.

6.3. Scale to Measure Perceived Control Over Technology

Two items were constructed to capture the autonomy that software professionals enjoy at work with respect to the choice of technology for any given project. The first item depicts the autonomy aspect of control. The second item captures the knowledge pertaining to decision making with regard to choice of technology. This encompasses the aspects of individual capability as well as availability of resources. Face validity and content validity have been incorporated in the questionnaire development process by using expert opinions.

7. DATA COLLECTION

This study adopted a cross-sectional design and survey methodology. Respondents to the study were software professionals who develop customized software for end users. They were located in two cosmopolitan cities (Chennai and

Bangalore) in South India. Convenient sampling was used in the study because of resource constraints and operational constraints. Personal interviews and e-mail were used to obtain responses. A total of 700 questionnaires were distributed, and 156 valid responses were obtained. Twenty-two software companies participated in the study. They included large multinational software companies as well as medium and small software companies.

7.1. Sample Profile

Analysis of the sample revealed the following key features. Eighty percent of the respondents were men. Eighty-four percent of the respondents were 22 to 30 years old. The mean age of the respondents fell between 26 and 30 years. Thirty-one percent of the respondents were married, and the rest were unmarried. Seventy-one percent of the respondents had undergraduate qualification in engineering. Fifty-three percent of the respondents had undergone software-training courses from private institutes. Seventy-five percent of the respondents work more than 8 hr a day.

8. STATISTICAL ANALYSIS

8.1. Factor Analysis and Reliability of the HCI Variables

Factor analysis has been conducted for the scales of CSE, technology-training efficacy, and perceived control over technology to determine their unidimensionality. The 12 items of the modified CSE scale were factor analyzed by using principal component method and varimax rotation to arrive at the loadings. The 12 items loaded as two factors. One set of items that pertained to situations that do not involve any form of help was termed *Self-Dependent CSE* (with a reliability of α = 0.85), and the other set of items that deal with situations that involve seeking help from various sources was termed as *Support-Dependent CSE* (with a reliability of α = 0.91; see Table 1). These two factors explain 71% of the total variance. From a conceptual framework, these two factors are two ends of the continuum. At the lower end of the continuum, there are software professionals who are confident of their computer skills and abilities if they know that there is help available in some form, either as a help desk that will guide them step by step or a person who will guide them. On the higher end of the continuum are those software professionals who are confident of their computer skills and who can work on any software technology without any help or support. Both contexts deal with confidence in their computing abilities from a conceptual angle, but the essential difference exists in the causal condition of whether the confidence is intrinsically there or whether it because of the availability of help.

Factor analyses indicate that both variables, perceived control over technology and technology-training efficacy, are unidimensional in nature. Cronbach's alpha is 0.6 in both cases. Factor analysis revealed that the scale of intrinsic motivation is also unidimensional. Its reliability was found to be 0.8.

Table 1: Factor Loadings—Computer Self-Efficacy

Preliminary Statement: *I could complete the program /design using a new software technology/ package/ language…*

Items	*Factor Loadings*
Factor 1: Support-dependent computer self-efficacy (Variance = 42.3%, α = 0.91)	
…if there was someone giving me step by step instructions.	.878
…if someone demonstrated how to do it first, before I try it.	.798
…and I will probably excel at it soon.	.766
…if I had used similar technologies before this one to do the same job.	.752
…if I had just the built-in help facility for assistance.	.743
…if someone else had helped me get started.	.729
…if I had a lot of time to complete the job for which the technology was provided.	.718
…if I could call someone for help if I got stuck.	.646
…if I had seen someone else using it, in the past before trying it myself.	.545
Factor 2: Self-dependent computer self-efficacy (Variance = 29.59%, α = 0.85)	
…if I had never used a technology like it before.	.872
…if there was no one around to tell me what to do as I go.	.830
…if I had only technology manuals for reference.	.799

8.2. Descriptives and Correlation Analysis

Descriptive statistics revealed that intrinsic motivation, technology-training efficacy and support-dependent CSE were high among software professionals (See Table 2). The mean values of self-dependent CSE and perceived control over technology were not as high as the other HCI variables.

Bivariate correlation was conducted to ascertain the relationship between stress, HCI variables, and outcome variables (See Table 3). Results indicated that:

- Stress was significantly negatively related to self-dependent CSE and support-dependent CSE. None of the other HCI variables was related to stress.
- Self-dependent CSE was significantly positively related to the other HCI variables such as intrinsic motivation, technology-training efficacy, and control over technology.
- Support-dependent CSE was positively related to intrinsic motivation and technology-training efficacy, but it was not related to control over technology.
- Technology-training efficacy and perceived control over technology were significantly positively related to intrinsic motivation

The results revealed that hypothesis (H1a) was partially supported. HCI variables were negatively related to stress. However, the relationship was significant only in the case of self-dependent CSE and support-dependent CSE. The results suggest that intrinsic motivation and technology-training efficacy can help in the development of support-dependent CSE. In addition, support-dependent CSE, in combination with perceived control over technology, can establish self-dependent CSE. There is a significant negative relationship between self-dependent CSE and work exhaustion. Challenging environment and innovative projects help software professionals to de-

Table 2: Measures of Central Tendency and Dispersion of Human-Computer Interaction Variables

Human-Computer Interaction Variables	*Sample Items: Characterize Yourself When You Use The Computer*	*Mean*	*SD*	*Range*
Intrinsic motivation	I am spontaneous, unimaginative, flexible, creative, playful.	5.116	0.823	3.28
Technology training efficacy	I am confident that I am able to train myself well in all the new technologies.	5.388	0.854	4.33
Perceived control over technology	I have control over the using any software technology/ platform that I think is appropriate for a given project.	4.512	1.180	6.00
Self-dependent computer self-efficacy	I could complete the program/design using new software … even if I had never used a technology like that before.	4.254	1.154	5.66
Support-dependent computer self-efficacy	I could complete the program/design using a new software … if someone demonstrated how to do it first, before I try it.	5.519	1.013	5.00

Table 3: Relationship Between Occupational Stress, Work Exhaustion, and Human-Computer Interaction Variables

	Stress	*WE*		*SuCSE*	*IM*	*TTE*	*CoT*
Stress	1						
WE	0.457***	1					
SCSE	–0.174*	–0.168*	1				
SuCSE	–0.168*	–0.065	0.659***	1			
IM	–0.070	–0.112	0.600***	0.59***	1		
TTE	–0.188	–0.073	0.371**	0.422***	0.459***	1	
CoT	–0.038	–0.008	0.245**	0.095	0.372***		1

***p < .001. **p < .01. *p < .05.

Note. WE = work exhaustion; SCSE = self-dependent computer self-efficacy; SuCSE = support-dependent computer self-efficacy; IM = intrinsic motivation; TTE = technology training efficacy; CoT = control over technology.

velop self-dependent CSE beliefs by providing adequate opportunities to control technology. Self-dependent CSE can be strengthened among software professionals by motivating them to acquire necessary training and enabling them to take control over technology so that they can prevent work exhaustion.

8.3. Hierarchical Moderated Multiple Regression

Hierarchical multiple moderated regression was performed to identify the possible additive and interaction effects of stress, self-dependent CSE, control over technology, and technology-training efficacy on work exhaustion. The concept of support-dependent CSE is significantly correlated with the concept of self-dependent CSE (Pearson's correlation coefficient is 0.659 at $p < .000$). Moreover, intrinsic motivation is also significantly correlated with self-dependent CSE (Pearson's correlation coefficient is 0.6 at $p < .000$) as well as support-dependent CSE (Pearson's correlation coefficient is 0.59 at $p < .000$). Hence, only self-dependent CSE is used in the hierarchical moderated regression analysis along with technology-training efficacy and perceived control over technology.

Predictors and moderators were jointly entered in Step 1, and the interaction terms were entered in Step 2. Regression coefficients for the main effects of predictor variables were obtained in Step 1 of each analysis, whereas the coefficients for interaction terms were obtained in Step 2. *F* values illustrate the significance of incremental R^2 at each step of the equation (Cohen & Cohen, 1983).

9. RESULTS AND DISCUSSION

To test the hypotheses H2a and H2b, hierarchical multiple moderated regression was performed, with work exhaustion as the dependent variable. The results of the regression are given in Table 4. A significant *F* value was found in the cases of both main (additive) effects as well as interaction effects. Despite the *F* values being significant, the incremental change in F value corresponding to the increase in R^2 value is not significant. Hence the data supported hypothesis H2a and did not support hypothesis H2b.

The results in Table 4 indicate that there was a significant relationship between occupational stress and work exhaustion. The regression coefficient of the significant interaction effect of stress × control over technology revealed a negative sign. Therefore, for lower values of control over technology, work exhaustion will be higher (by signed coefficient rule of Mossholder, Kemery, & Bedeian, 1990). This gives an important finding that if control over technology can be enhanced, work exhaustion can be reduced for software professionals.

Table 4: Human-Computer Interaction Variables as Moderators in the Relationship Between Stress and Work Exhaustion

Predictors	R^2	*F*	Δ^2	*Significance of F Change*	*B Value*	*Significance*
Model 1						
Constant	0.220	10.644***	0.219	0.000	–0.074	–.097
Stress				0.672***	6.067	
Control over technology				0.026	.315	
Technology training efficacy				0.070	.583	
Self-dependent computer self-efficacy				–0.126	–1.434	
Model 2						
Constant	0.251	7.103***	0.031	0.105	–2.10	–.943
Stress				1.517	1.909	
Control over technology				0.643*	2.300	
Technology training efficacy				–0.144	–.312	
Self-dependent computer self-efficacy				–0.016	–.058	
Stress × Technology training efficacy				0.080	.492	
Stress × Control over technology				–0.238*	–2.330	
Stress × Self-dependent computer self-efficacy				–0.053	–.507	

Note. *** $p < .001$. ** $p < .01$. * $p < .05$.

Only a few studies have explored the moderating role of computer self-efficacy in the relationship between stress and burnout (e.g., Salanova et al., 2002). These studies dealt with users of information technology and not with software professionals. This study is perhaps the first with respect to software professionals in which the variables of self-dependent CSE, technology-training efficacy, and control over technology have been concomitantly considered. Confidence in training leads to confidence in performing software development by oneself, which is due to control over technology gained in the process of software development.

The results imply that HCI variables have main effects but no interaction (moderating) effects in the relationship between occupational stress and work exhaustion. Control over technology has significant interaction effect in the relationship between stress and work exhaustion as can be seen in Model 2 (Table 4). Hence, efforts can be taken to empower software professionals with adequate control over technology to prevent work exhaustion.

9.1. Implications

Good HCI personality enables people to effectively deal with stress. In particular, control over technology has emerged as a key variable that can play a critical role in enabling people to deal with stress in a better manner. The important aspect with respect to control over technology is the requisite knowledge that enables software professionals to make appropriate decisions regarding the choice of technology for various uses. Self-dependent CSE, intrinsic motivation, and technology-training efficacy influence control over technology. Creating a favorable environment for experimentation and nurturing creativity and innovation in the design and use of technology enhance intrinsic motivation and self-dependent CSE. Therefore, enhancing the earlier-mentioned aspects would empower software professionals with control over technology. To achieve this, software companies must provide adequate information and resources for learning. Further, organizations can provide opportunities for peer learning as well as encourage communities of practice for sharing knowledge. These communities provide stability and security in the face of changing technological environment and enable software professionals to perceive greater control over technology.

9.2. Conclusion

The objective of this article was to assess whether HCI factors moderate the relationship between occupational stress and work exhaustion. In this process, this article also dealt with the modification of the CSE scale and the development of scales to measure technology-training efficacy and perceived control over technology in the context of software professionals. The results indicate that HCI factors have main effects in the relationship between stress and work exhaustion but do not moderate the relationship. The findings suggest that training in technical skills, intrinsic motivation at work, and confidence in the computing abilities of oneself can increase the perceived control over technology that will mitigate the negative consequences of stress, particularly work exhaustion.

9.3. Future Directions

There is scope for further work in this area. It is imperative to explore whether HCI factors vary across cultures, particularly in the context of global work culture. In addition, it would be interesting to investigate the applicability of this work in various kinds of projects such as time and maintenance projects and turnkey projects. Further, it would be pertinent to explore whether there are differences in the moderating effect of HCI variables between offshore and on-site teams. It also would be valuable, particularly from the industry perspective, to determine whether there are differences in the moderation of HCI variables depending on the phases of the software development lifecycle. In addition, it may be essential to explore whether experience in the industry has a differentiating effect on the model proposed in the study. This study has only explored only the tip of the iceberg regarding the role of HCI variables in the realm of occupational stress, and there is ample scope to develop this model both from the perspective of academicians and practitioners in the industry to increase the well-being of software professionals.

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